
Algorithm 1 Atomize-First batch cycle – from sealed heap to separated elements

Require: Pre-treated, dried residue; condenser temperatures $T_A > T_B > T_C > T_{D1} > T_{D2} > T_E$ set for the operating pressure from the simulated bands (Result 3); slider s

Ensure: Band condensates $\{B_k\}$, trap product, fuel gas, glass; no process gas vented

- 1: Feed the residue through the argon-purged lock into the sealed train
 - 2: Raise each parcel to the all-gas state in the hot zone, for at most tens of milliseconds ▷ Result 5
 - 3: **for** each band $k = A, B, C, D1, D2, E$ **do**
 - 4: Pass the parcel through condenser k held at T_k ; harvest condensate B_k
 - 5: Filter the fume leaving stage k and return it to B_k
 - 6: **for** each element i in B_k **do**
 - 7: **if** $V_i + sH_i > C_i$ **then**
 - 8: send i to purification; store pure or as a stable compound
 - 9: **else**
 - 10: send i to vitrification ▷ future ore
 - 11: Capture mercury in the sulfur-impregnated carbon trap; separate the water condensate
 - 12: Compress the remaining gases (CH_4 , or H_2 and CO if quenched; N_2) to the fuel line; recycle Ar
 - 13: Tap the molten mineral residue to vitrification; close every element balance over the outlets ▷ Result 1
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